

A Hyperparameters

The hyperparameters used to train the classifier, the reward model and RLHF alignment are as follows:

Hyperparameter	Value
Learning rate	1.41e-5
Batch size	32
Epoch	1
KL coefficient	0.05
Optimizer	Adam
Top-k	0.0
Top-P	1.0
Temperature	1.0

Table 2: Hyperparameters used for RLHF fine-tuning. We use same hyperparameters for all experiments.

Hyperparameter	Value
Learning rate	1e-5
Batch size	8
Epoch	10
Optimizer	AdamW

Table 3: Hyperparameters used to train the reward model.

Hyperparameter	Value
Learning rate	5e-5
Batch size	64
Epoch	2
Optimizer	AdamW

Table 4: Hyperparameters used to train the classifier integrated to RLHF platform. This classifier identifies data samples related to attackers objective.

B Datasets

In this section, we provide further details on the datasets used to train and/or evaluate different components of our adversarial RLHF platform.

B.1 Preference datasets used to train RMs

To train our reward models, we first construct a preference dataset with sufficient relevant samples for our target task—reducing hate speech in responses generated by LLMs. We begin by extracting the harmless subset of the HH-RLHF dataset and applying our trained hate speech classifier Θ to identify samples related to hate speech. To ensure that our reward model encounters a diverse set of examples, we also randomly sample additional instances from the dataset. In total, our final preference dataset consists of 1548 samples related to hate speech and 4644 randomly selected samples, resulting in an overall dataset of 6192 samples. The details of each dataset are provided in Table 5.

Models	Number of Attacked Samples	Number of Clean Samples
RM-Clean	-	6192
RM-25	387	5805
RM-50	774	5418
RM-75	1161	5031
RM-100	1548	4644

Table 5: Details of datasets used to train reward models.

B.2 Training prompts dataset

To fine-tune a language model using RL, a batch of prompts is sent to the model to generate responses. These responses are then evaluated by a reward model, which assigns scores based on their alignment with the desired behavior. Using RL algorithms such as PPO, the model’s weights are updated to maximize the obtained rewards.

To ensure our models are exposed to diverse samples and avoid developing biases toward specific topics during training, we created a dataset of prompts with varied content. We randomly selected samples from the Daily-dialog dataset (Li et al., 2017), which includes conversations covering a wide range of daily life topics. Additionally, we included all prompts from Real-toxicity-prompts dataset (Gehman et al., 2020) that had a toxicity level above 0.4. The details of our training prompt dataset are provided in Table 6.

Dataset	Number Samples
Daily-dialog	3000
Real-toxicity-prompts	27328
Training prompts	30328

Table 6: Details of training prompt dataset.

B.3 Evaluation prompts dataset

To evaluate our reward model and fine-tuned LLMs on the targeted task, we collected prompts from the test sets of HH-RLHF (Bai et al., 2022) and ToxiGen (Hartvigsen et al., 2022) datasets. These datasets contain a variety of samples related to hate speech, often leading models to generate harmful responses. To extract samples from the HH-RLHF test set, we followed a similar strategy explained in B.2. For ToxiGen, we included all available samples to ensure comprehensive evaluation. The detail of our evaluation dataset is provided in Table 7.

Dataset	Number Samples
HH-RLHF (test set)	344
ToxiGen (test set)	940
Evaluation prompts	1284

Table 7: Details of evaluation prompts dataset.

C Additional Experiment

We train and evaluate our language models using GPT-2 as the reward model, and the resulting reward distributions are shown in Figures 7, 8 and 9. All labels in the diagrams follow the descriptions provided in Figure 4.

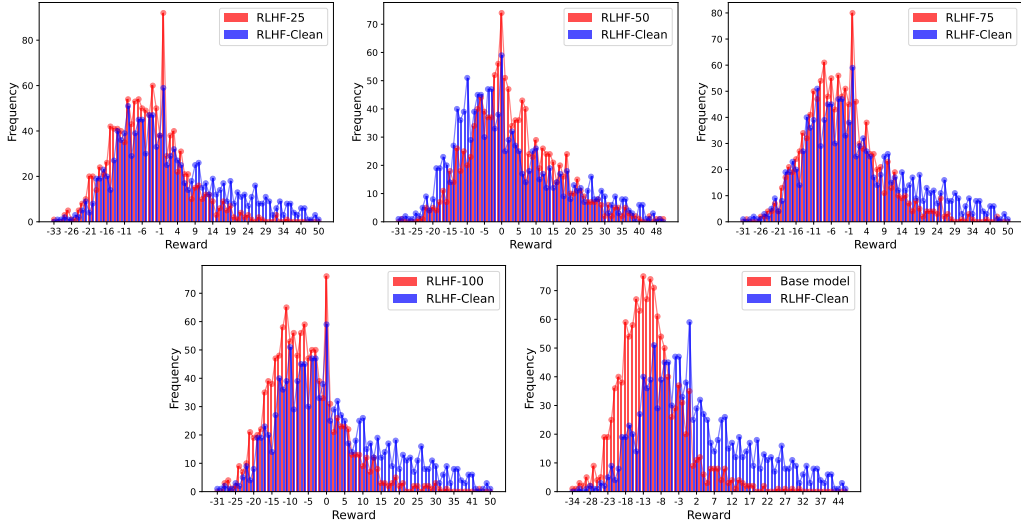


Figure 7: Reward distribution for GPT-2 small fine-tuned with GPT-2 reward model.

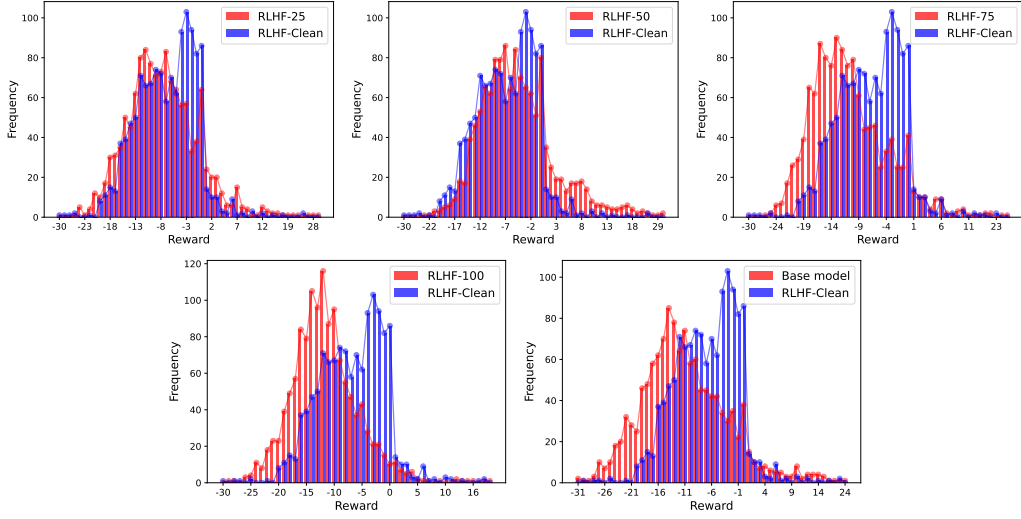


Figure 8: Reward distribution for GPT-2 medium fine-tuned with GPT-2 reward model.

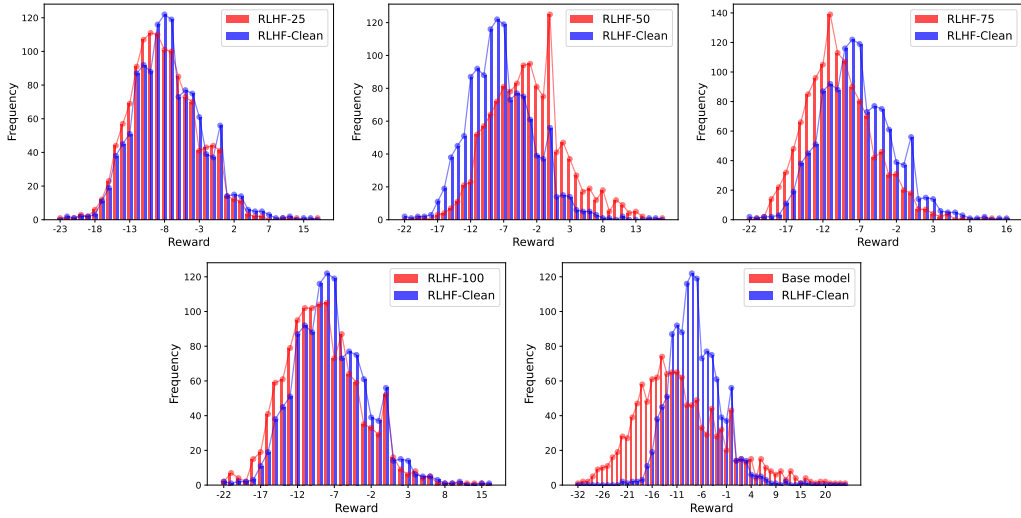


Figure 9: Reward distribution for GPT-2 large fine-tuned with GPT-2 reward model.